

The Execution Time of the "Compare-Exchange" Operation τ (sec):

Test number	The Size of the Data to Be Sorted	The Execution Time of Bubble Sorting (sec.)	The Execution Time of Sorting from the Standard Library (sec.)
1	10	0.000007	0.000005
2	100	0.000049	0.000012
3	10000	0.233521	0.000679
4	20000	0.898994	0.001457
5	30000	2.041505	0.002229
6	40000	3.638186	0.003625
7	50000	5.757199	0.006333

Test Number	The Size of Data to Be Sorted	Execution Time (sec)	Theoretical Time (sec)
1	10	0,000007	0,00000020
2	100	0,000049	0,00002247
3	10000	0,233521	0,22697730
4	20000	0,898994	0,90795460
5	30000	2,041505	2,04293190
6	40000	3,638186	3,63190920
7	50000	5,757199	5,67488650

$\tau=0.0000000454$

Test Number	Data amount	Serial Bubble Sorting	Serial Standart Sorting	Parallel Algorithm. The		
				number of processors		
				2	4	8
1	10	0,000007	0,000005	0,000634	0,000091	0,002401
2	100	0,000049	0,000012	0,000468	0,000108	0,00078
3	10000	0,233521	0,000679	0,03914	0,011704	0,00972
4	20000	0,898994	0,001457	0,180702	0,041866	0,024806
5	30000	2,041505	0,002229	0,497286	0,098568	0,049611
6	40000	3,638186	0,003625	0,937952	0,190542	0,087931
7	50000	5,757199	0,006333	1,535971	0,330815	0,10511

Test Number	Parallel Algorithm					
	2 Processors		4 Processors		8 Processors	
	Speed Up 1	Speed Up 2	Speed Up 1	Speed Up 2	Speed Up 1	Speed Up 2
1	0,011041	0,007886	0,076923	0,054945	0,002915	0,002082
2	0,104701	0,025641	0,453704	0,111111	0,062821	0,015385
3	5,966300	0,017348	19,952239	0,058014	24,024794	0,069856
4	4,975009	0,008063	21,473129	0,034802	36,240990	0,058736
5	4,105294	0,004482	20,711641	0,022614	41,150249	0,044930
6	3,878862	0,003865	19,093880	0,019025	41,375465	0,041226
7	3,748247	0,004123	17,403077	0,019144	54,773085	0,060251